

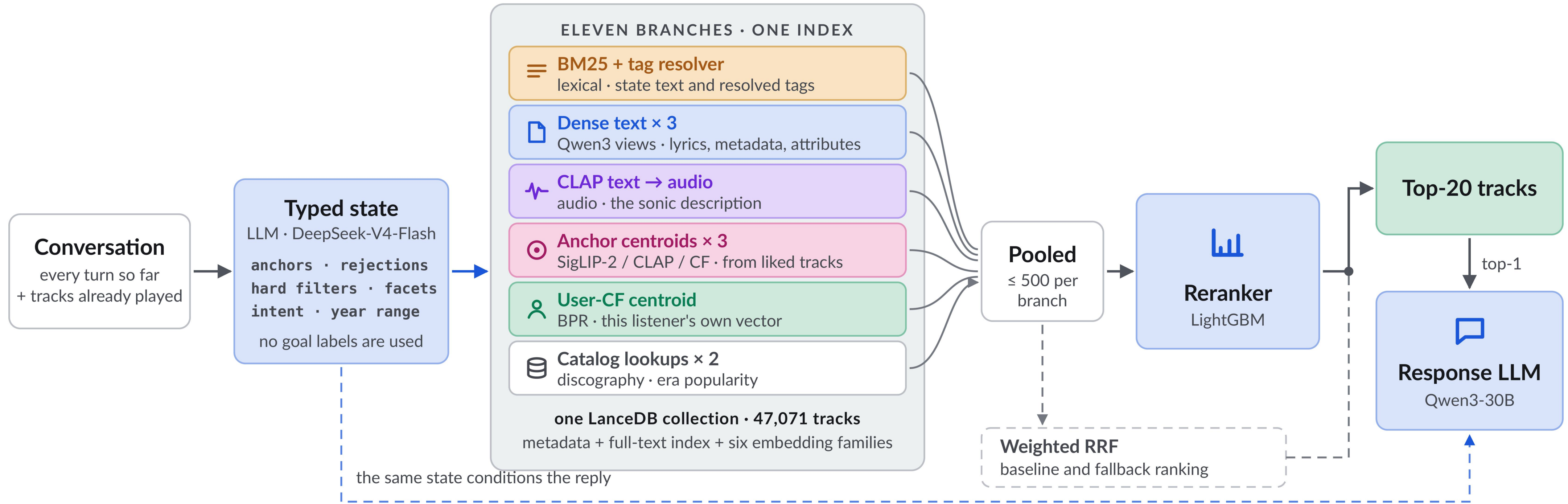
State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

Nidhin Pattaniyil · Semih Yagli · Tanwir Zaman
Independent Researchers, Hoboken, New Jersey, USA

11 retrieval branches
146 ranker features
29/40 where that finished



Paper, poster, code, data
npatta01.github.io/music-crs-2026



What the system reads off a conversation

Earlier turns: we played Neko Case’s “Calling Cards” (“*not quite the one*”), then “Man” (“*not it either*”).

User, now: “...much more subdued... a specific sense of place in the lyrics... a somewhat stark, almost a cappella delivery.”

EXTRACTED STATE

anchor artist Neko Case (must use) · earlier turns
rejected “Calling Cards”, “Man” · earlier turns
facets mood = subdued
theme = sense of place
sonic = stark, almost a cappella
request hidden_target

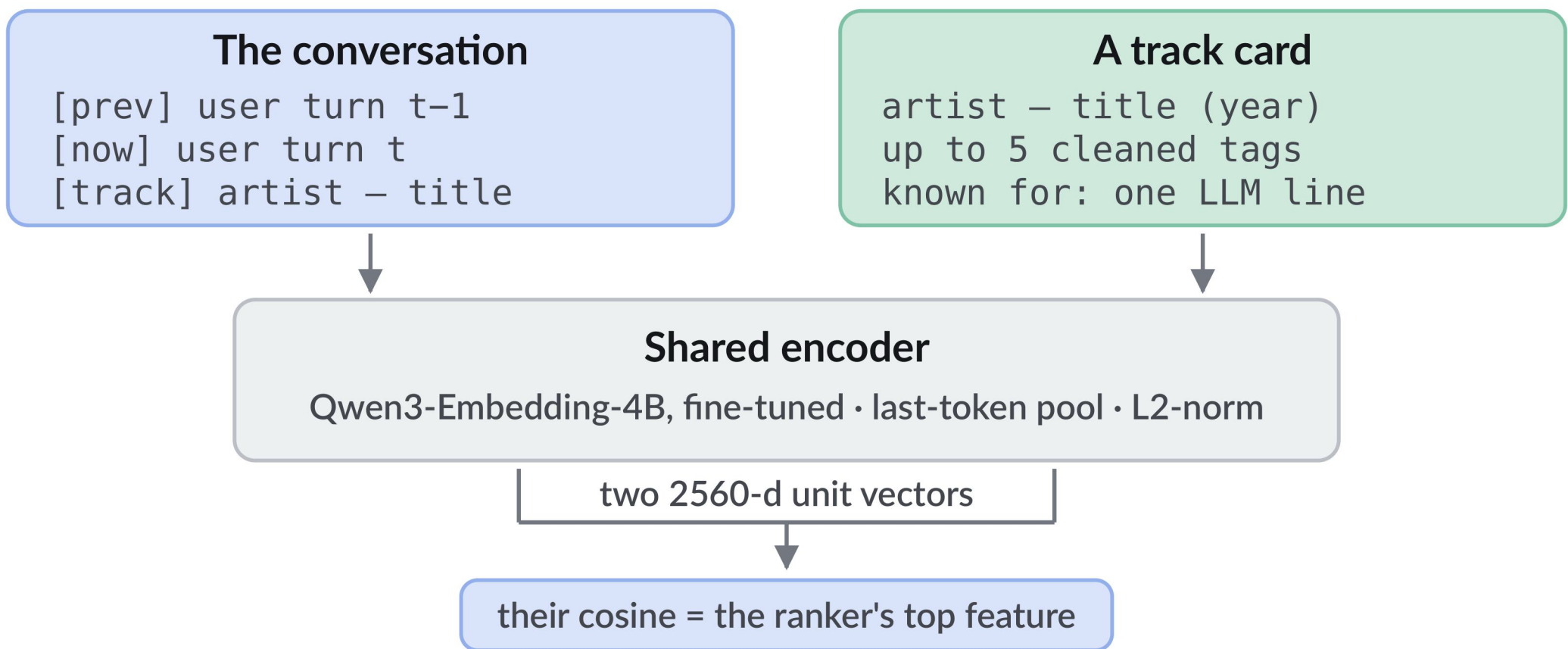
The full schema: request type · intent mode · anchors and pinned tracks · per-track feedback · rejected tracks and artists · hard filters · mood, genre and sonic facets · process constraints · routing flags · lyrical theme · release-year window.

What the ranker leans on

FEATURE FAMILY	SHARE OF GAIN
Bi-encoder cosine	59.0%
Per-branch rank and score	11.5%
Session and artist affinity	10.7%
Popularity · state · tags	9.2%
Other similarity cosines	8.9%
Constraints and rejections	0.7%

LightGBM LambdaMART · 146 features per (turn, track) pair.

THE BI-ENCODER BEHIND THAT TOP FEATURE



How it scored

OFFICIAL SPLIT	NDCG@20	JUDGE	COMPOSITE
Blind-A · development phase	0.4380	4.70	0.5799
Blind-B · final phase	0.2537	3.30	0.3811

What is not ideal

We trained and validated the ranker on the same dev turns.

THE LISTENER ASKED FOR

OUR TOP PICK

We extracted more than we used

the Kamelot track
from Silverthorn

Kamelot — Fallen Star,
from Haven
We extracted “Silverthorn” but
never filtered on it

We used no data from outside the catalog

exactly 126.70 bpm,
in G minor

the catalog has no tempo or
key, and we added no
outside source

The reply said “Got it” for a track we did not have

“Watercolors”
by Pat Metheny

Metheny — Alfie, with the
reply “Got it — here’s ‘Alfie!’...”
“Watercolors” is not in the catalog

The training labels often ignore the request

User: “...similar chill vibe, but *from a different artist?*” (just played: Bonobo)

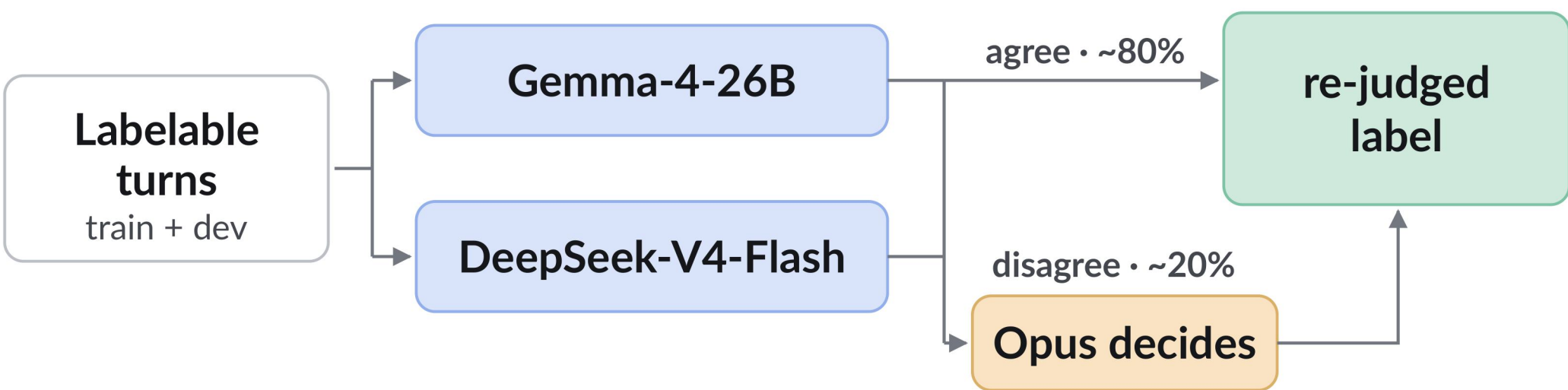
Target: **Bonobo — “Jets”** · annotated MOVES toward goal

Next turn: “...hoping for something *from a different artist this time...*”

Target: **Bonobo — “Pieces”** · annotated MOVES toward goal

The label is whatever played next in an LLM-generated session, even when the listener had just asked for something else.

OUR FIX: TWO LLM JUDGES RE-SCORE THE LABELS



Agreement split measured on the training turns.